

# Manganese

**25**  
**Mn** The Swedish chemist Johan Gottlieb Gahn was the first to isolate pure manganese in 1774. This element is essential in our diet in small quantities. Cereals, nuts, beets, and various other fruits and vegetables, such as blueberries, avocados, and olives, are all good sources.



Laboratory sample of pure manganese

*This silvery metal is dense and brittle.*



## Cave art

Many pigments used in ancient paintings, such as the Lascaux cave paintings in France, come from compounds containing manganese. Oxides of manganese tend to produce brown and black colors.

## Better gasoline

A complex compound of manganese, carbon, hydrogen, and oxygen—known as MMT—is added to gasoline in some countries, particularly the US, to make cars run more efficiently. But as it is toxic, it's being phased out. Almost no MMT is used in fuel in the European Union today.



*High-quality gas protects engines from wear.*

# Iron

**26**  
**Fe** Produced in huge quantities, iron is the most important metal used in the world today. While construction is its most common application, it is also used as a catalyst in many industrial processes, such as in the manufacture of ammonia for fertilizers. In human blood, iron is present in hemoglobin, a compound that carries oxygen around the body.



Chunk of pure iron refined in a laboratory



## Rust red

Iron corrodes in a natural chemical process called oxidation—rusting. When exposed to water and oxygen, its surface gets coated with a reddish-orange layer of hydrated iron oxide that cracks and flakes off, wearing away the metal.

## Iron power

When forged into weapons and machinery, iron brought economic and political power to many past empires. In the modern world, it is an important material in construction, often in the form of steel—an alloy of iron and carbon—but also as cast and wrought iron.

The Eiffel Tower, Paris, France

Wrought-iron lattice

